

Assignment 3

Name: _____

Student ID: _____

12 problems. Show all of your work.

Problem 1.

For approximately what values of x can we replace $\sin(x)$ by $x - \frac{x^3}{6}$ with an error of magnitude no greater than 4×10^{-3} ?

Problem 2.

Use the remainder estimation theorem to estimate the maximum error when approximating $f(x) = e^x$ by $T_2(x) = 1 + x + x^2$ on the interval $\left[-\frac{5}{6}, \frac{5}{6}\right]$.

Problem 3.

Consider the function $f(x) = \ln(1 + 2x)$.

- A. Find the second order Taylor polynomial $T_2(x)$ centered at 0.
- B. Estimate the maximum error when approximating $f(x) = \ln(1 + 2x)$ with the second order Taylor polynomial $T_2(x)$ centered at 0, on the interval $[-0.25, 0.25]$.

Problem 4. Approximating The Electric Field due to an Electric Dipole

Consider an electric dipole consisting of two charges, $+q$ and $-q$, separated by a distance d . The electric field at a point P located at a distance r from the positive charge along the axis of the dipole is given by:

$$E = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r^2} - \frac{q}{(r+d)^2} \right)$$

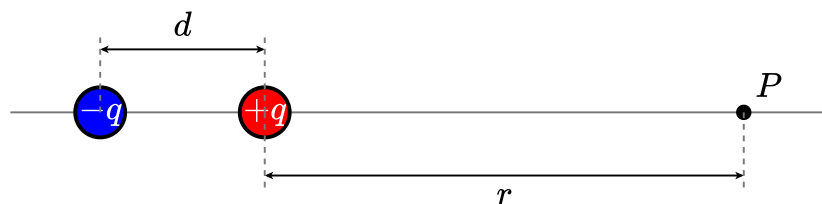


Figure 1. An electric dipole: charges $+q$ and $-q$ separated by a distance d , with the field point P a distance r from the positive charge along the axis.

Use the Taylor series to approximate the electric field E at point P for $d \ll r$. Show that the leading term in the approximation is proportional to d/r^3 .

Problem 5. *How small must the separation be?

Starred problem. This one is for the interested reader. It will not be examined.

A dipole is often treated as a **point dipole** by replacing the exact axial field

$$E = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r^2} - \frac{q}{(r+d)^2} \right)$$

with its leading-order Taylor term for $d \ll r$,

$$E \approx \frac{1}{4\pi\epsilon_0} \cdot \frac{2qd}{r^3}.$$

Suppose this approximation must agree with the exact field to within a relative error of **1%**. If the field point P is $r = 3.0 \text{ cm}$ from the positive charge, how small must the charge separation d be for the point-dipole approximation to be valid?

Problem 6.

Use a **binomial series** to represent the function $f(x) = \frac{2x}{1-x}$ as a series. Write your final answer in sigma notation.

Problem 7.

Use the binomial series $(1+x)^m = 1 + \sum_{k=1}^{\infty} \binom{m}{k} x^k$ to obtain the Maclaurin series for

A. $\frac{1}{(1+x)^3}$. Write your answer in sigma notation, $\sum_{k=0}^{\infty} (\dots)$.

B. $\sqrt[3]{1+x}$. Give the first four terms only.

Problem 8.

Expand the quantity $\sqrt[3]{P+t}$ about 0 in powers of $\frac{t}{P}$. Give four nonzero terms.

Problem 9.

Let

$$f(x) = 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \dots$$

Find the exact value of $\int_0^{1/13} f(x) dx$.

Problem 10.

Evaluate $\int \frac{\cos x - 1}{x} dx$ as an infinite series. Give the first three nonzero terms.

Problem 11.

Use Taylor series to evaluate

$$\lim_{x \rightarrow 0} \frac{\cos(2x) - 1}{x^2 e^x}.$$

Problem 12.

The gravitational force exerted by the Earth on an object of mass m at height h above the surface is

$$F = \frac{mgR^2}{(R+h)^2},$$

where R is the radius of the Earth and g is the acceleration due to gravity at the surface.

- Express F as a Maclaurin series in the variable h/R , writing out the first four non-zero terms.
- What approximation to F do you get by keeping only the first term? When is it reasonable?